# Robot Kinematics Analysis Report (MDH)

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## 1 Modified DH Parameters

The kinematic structure of the robot is defined by the following Modified DH parameters (Craig's Convention). The transformation matrix $T_i^{i-1}$ is calculated as:

$$T_i^{i-1} = Rot_x(\alpha_{i-1})Trans_x(a_{i-1})Rot_z(\theta_i)Trans_z(d_i) \tag{1}$$

Table 1: MDH Parameters Table

| Joint $i$ | $\alpha_{i-1}(^\circ)$ | $a_{i-1}$ (mm) | $d_i$ (mm) | $\theta_{offset}(^\circ)$ |
|---|---|---|---|---|
| 1 | 0.0 | 40.0 | 0.0 | 0.0 |
| 2 | 90.0 | 80.0 | 100.0 | 0.0 |
| 3 | 90.0 | 0.0 | 0.0 | 180.0 |
| 4 | 90.0 | 100.0 | 0.0 | 0.0 |
| 5 | 90.0 | 100.0 | 0.0 | 0.0 |
| 6 | 90.0 | 0.0 | 100.0 | 180.0 |
| 7 | 90.0 | 80.0 | 0.0 | 0.0 |

## 2 Kinematics Visualization

The following figure illustrates the initial configuration and the coordinate frames of each joint.

## 3 Fourier Series Trajectory Representation

For smooth trajectory planning, the joint position $q(t)$ is represented using a finite Fourier series:

$$q(t) = a_0 + \sum_{n=1}^{N} \left( a_n \cos(n\omega_f t) + b_n \sin(n\omega_f t) \right) \tag{2}$$

Where:

- $a_0$: DC component (average position).
- $a_n, b_n$: Fourier coefficients.
- $\omega_f$: Fundamental frequency of the motion.

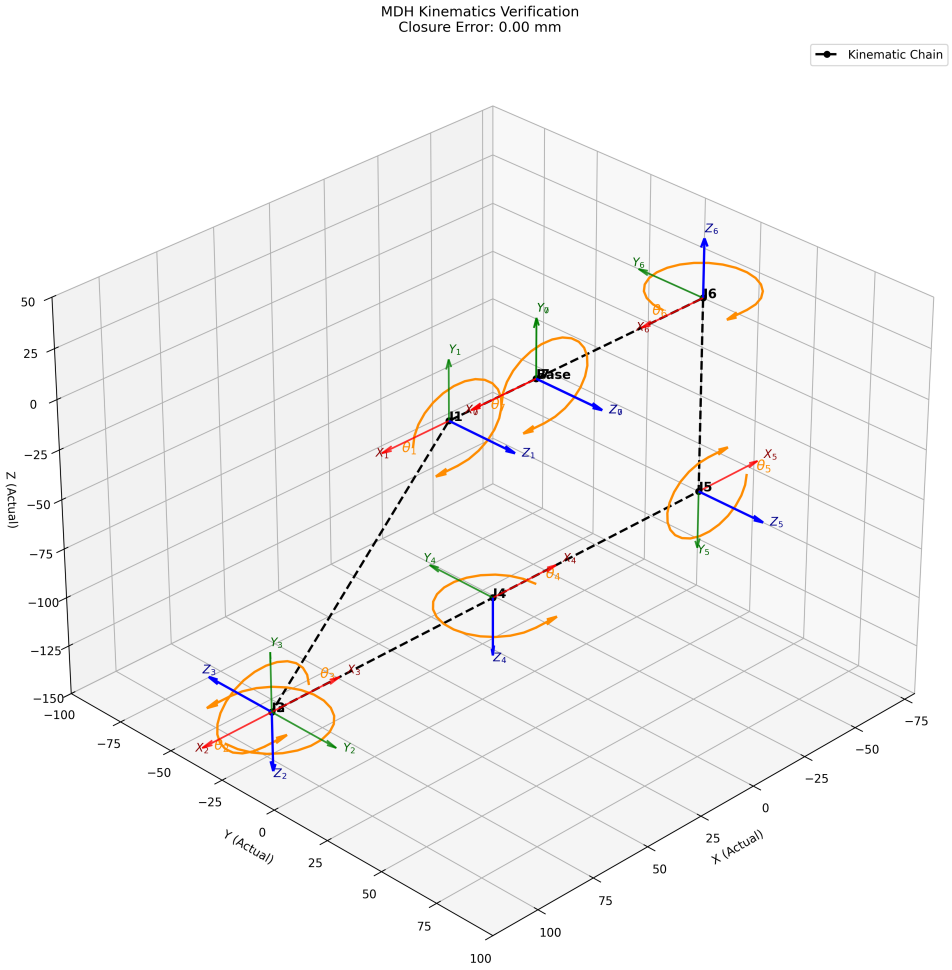



Figure 1: 3D Kinematic Chain Visualization